

Manoj Arulmurugan

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EDUCATION

MS, Data Science, University of Wisconsin-Madison ☑ Sep 2024 – May 2026 | Madison, WI
Statistical Modeling, Optimization, NLP, Advanced Deep Learning (GPA: 3.84/4.0)

Graduate Certificate, Data Science and Applications, IIT Madras ☑ Jun 2023 – Dec 2023

B.Tech, Computer Science, Vellore Institute of Technology GPA: 8.96/10 ☑ Jul 2019 – May 2023 | Chennai, India

EXPERIENCE

Machine Learning Intern, Calix, Inc. Jun 2025 – Dec 2025 | San Jose, CA, USA

- Architected a centralized **Feature Store** on Snowflake integrating **10+** data sources at account-week granularity, engineering **100+** features to standardize model inputs and eliminate redundant query development across the team.
- Redesigned the production Account-Level Regression model end-to-end, building **13 week-specific models** with algorithm-specific feature selection (**VIF + Genetic Algorithm** for linear; **SHAP** for tree models), achieving **~50% MAPE reduction**.
- Productionized the full pipeline on **Snowflake-dbt**, reducing dimensionality by **~80%** through Genetic Algorithm-based feature selection and optimizing the training window to meet the 30-minute warehouse timeout and reduce compute costs.
- Contributed to the broader **forecasting ensemble** (SARIMA + LSTM + ALR) achieving **~60%** overall accuracy improvement.
- Collaborated with Business Analysts and Data Engineers on **cohort-based account analysis** tracking revenue expansion and churn trends across customer acquisition, growth, and churn stages.

Operations Data Analyst, Shell India Market Pvt. Ltd. ☑ Aug 2023 – Aug 2024 | Bangalore, India

- Owned analytics for B2C apps, building **PowerBI** dashboards querying a **10TB data lake** via **SQL** across ServiceNow, Jira, and SharePoint; automated ETL pipelines, accelerating processing 3x and driving **\$250k in cost savings**.
- Designed KPI tracking framework adopted as primary analytics tool across B2C and B2B teams at Shell India; recognized with the **Shell Mobility Wall of Fame Award (Q1 2024)** ☑ for delivering high-impact data solutions.

Data Science Intern, Sedin Technologies Pvt. Ltd. ☑ Sep 2022 – Feb 2023 | Chennai, India

- Built a two-stage ML forecasting pipeline for an Indian fintech NBFC using **MLForecast**, engineering features across **400+** customer/region/risk combinations and applying clustering to reduce dimensionality to 6 behavioral groups.
- Improved forecast accuracy from **76% to 95%** using a **RandomForest + XGBoost** ensemble (SMAPE 4.5%, R^2 of 0.99), supporting 3-month loan disbursement forecasts for resource allocation and budgeting decisions.

PROJECTS

Production-Scale GRU4Rec: E-Commerce Recommendation System ☑, Big Data · GCP · MLOps

- Processed **280M raw events** (**BigQuery** → **PySpark** on Dataproc); applied bot removal, deduplication, and session segmentation to produce **2.88M** training sessions; trained GRU4Rec with event-type features and full softmax over 222K items.
- Achieved **NDCG@20=0.26** and **HR@20=0.48**, exceeding **T4Rec** XLNet+RTD by +5.1%; scaled to 1M users on **Vertex AI (A100)**.
- Deployed FastAPI + FAISS on **Cloud Run** (<500ms); built end-to-end MLOps loop: Jensen-Shannon divergence **concept drift detection** triggers automated Cloud Scheduler **model retraining** on Vertex AI; experiments tracked with **MLflow**.

Customer Churn Prediction, Non-Contractual Setting ☑, Time-Series · Statistical Modeling · ML

- Engineered **dynamic churn labels** on a real enterprise sales dataset using per customer-product **Empirical Cumulative Distribution Function** thresholds across **370k** records, capturing purchase-gap heterogeneity without fixed cutoffs.
- Benchmarked forecasting-based, statistical, and ML-based approaches; **LightGBM** and **RandomForest** won at **87%** accuracy.

Time-Aware RAG: Temporal Retrieval + Fast MRAG Re-Ranking ☑, Retrieval-Augmented Generation (RAG)

- Developed end-to-end temporal RAG pipeline: generated **15k synthetic training pairs via T5** on FineWeb, then contrastively **fine-tuned Contriever** with temporal hard negatives and MS MARCO regularization to prevent catastrophic forgetting.
- Pre-computed Sliding-Window MaxSim re-ranker replaces LLM summarization; ablation shows independent additive gains.
- Added **grounded LLM generation** with temporal citation prompting; **Hit@1 59% vs 40% on CAQA**; deployed to **HuggingFace**

SquadPlanner: AI-powered Trip Planning Application ☑, LangGraph · Multi-Agent Systems · MongoDB

- Architected a stateful multi-agent LLM system in **LangGraph**; MongoDBCheckpointner **serializes TripState** across HTTP for **human-in-the-loop** approval; post-gen edits re-enter graph at the affected node via `astream_events`, skipping upstream.
- Parsed free-text into typed hard constraints gating tool selection; streamed **real-time agent progress** via SSE.

SKILLS

Languages & Frameworks: Python, R, SQL, Julia, PyTorch, PySpark, Apache Spark, XGBoost, Hugging Face Transformers

Cloud & MLOps: GCP (BigQuery, Dataproc, Cloud Run, GCS), Snowflake, MLflow, Docker, FastAPI, dbt, Git

Data & Tools: LangChain, LangGraph, Darts, Pandas, NumPy, Polars, Feature Stores, ETL, Tableau, PowerBI

ML & Statistics: Forecasting, MLOps, GenAI, LLMs, RAG, Transformers, Multi-Agent Systems, A/B & Hypothesis Testing